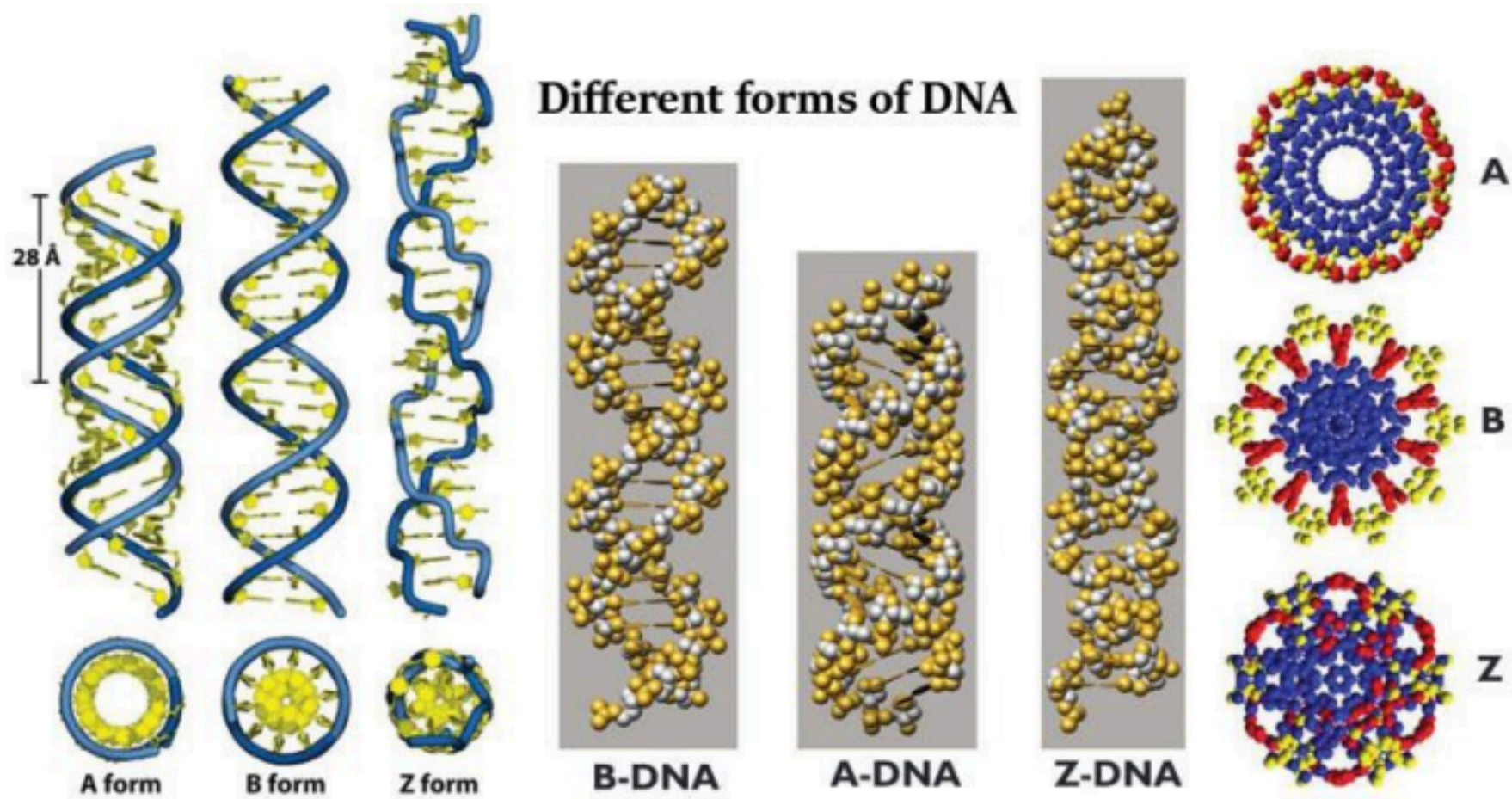




<https://microbenotes.com/different-forms-of-dna-b-form-a-form-z-form/#different-forms-of-dna-a-form-b-form-z-form>

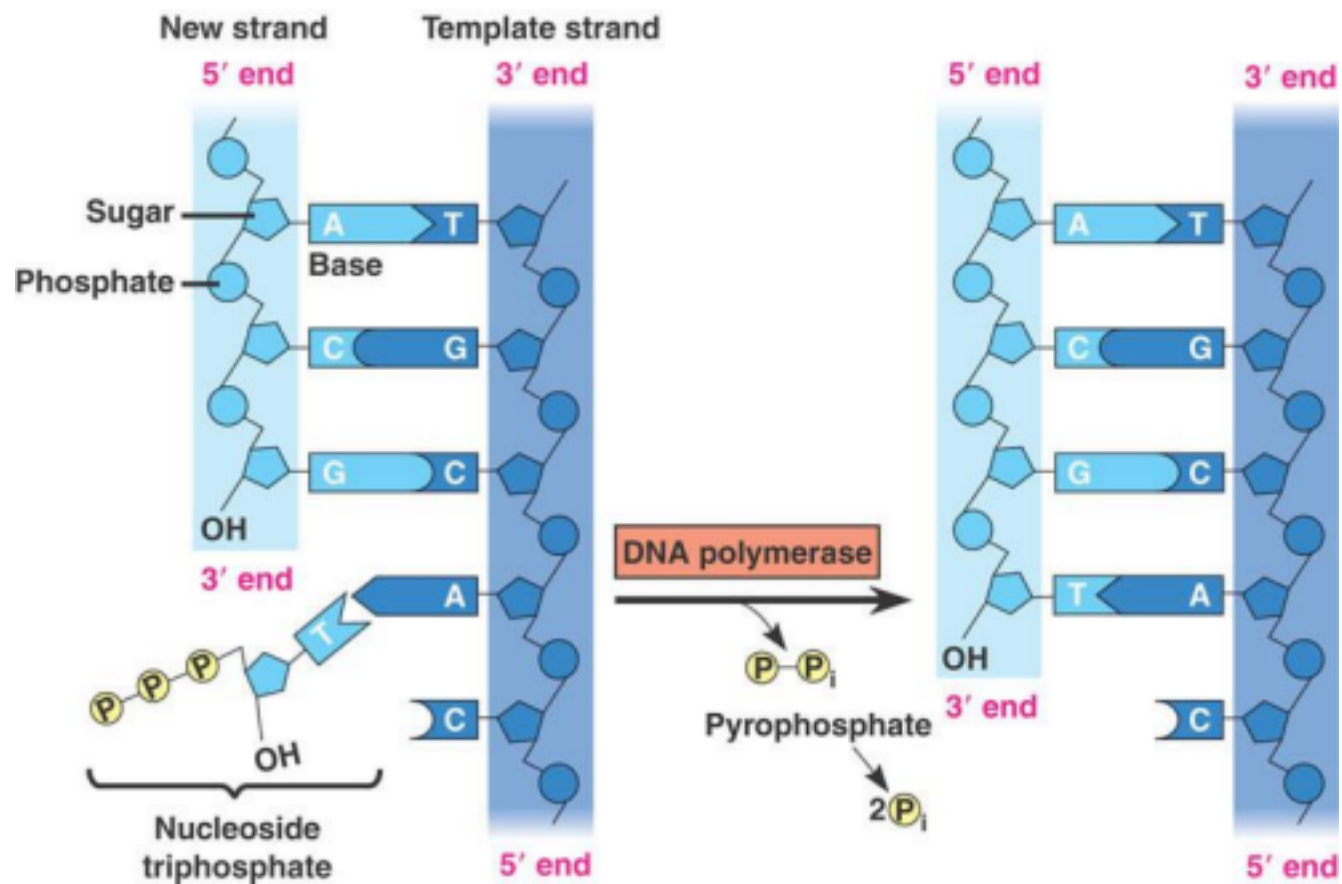
Table 2.5.12.5.1 Comparisons of B-form, A-form and Z-DNA			
	B-Form	A-Form	Z-Form

helix sense	Right Handed	Right Handed	Left Handed
base pairs per turn	10	11	12
vertical rise per bp	3.4 Å	2.56 Å	19 Å
rotation per bp	+36°	+33°	-30°
helical diameter	19 Å	19 Å	19 Å

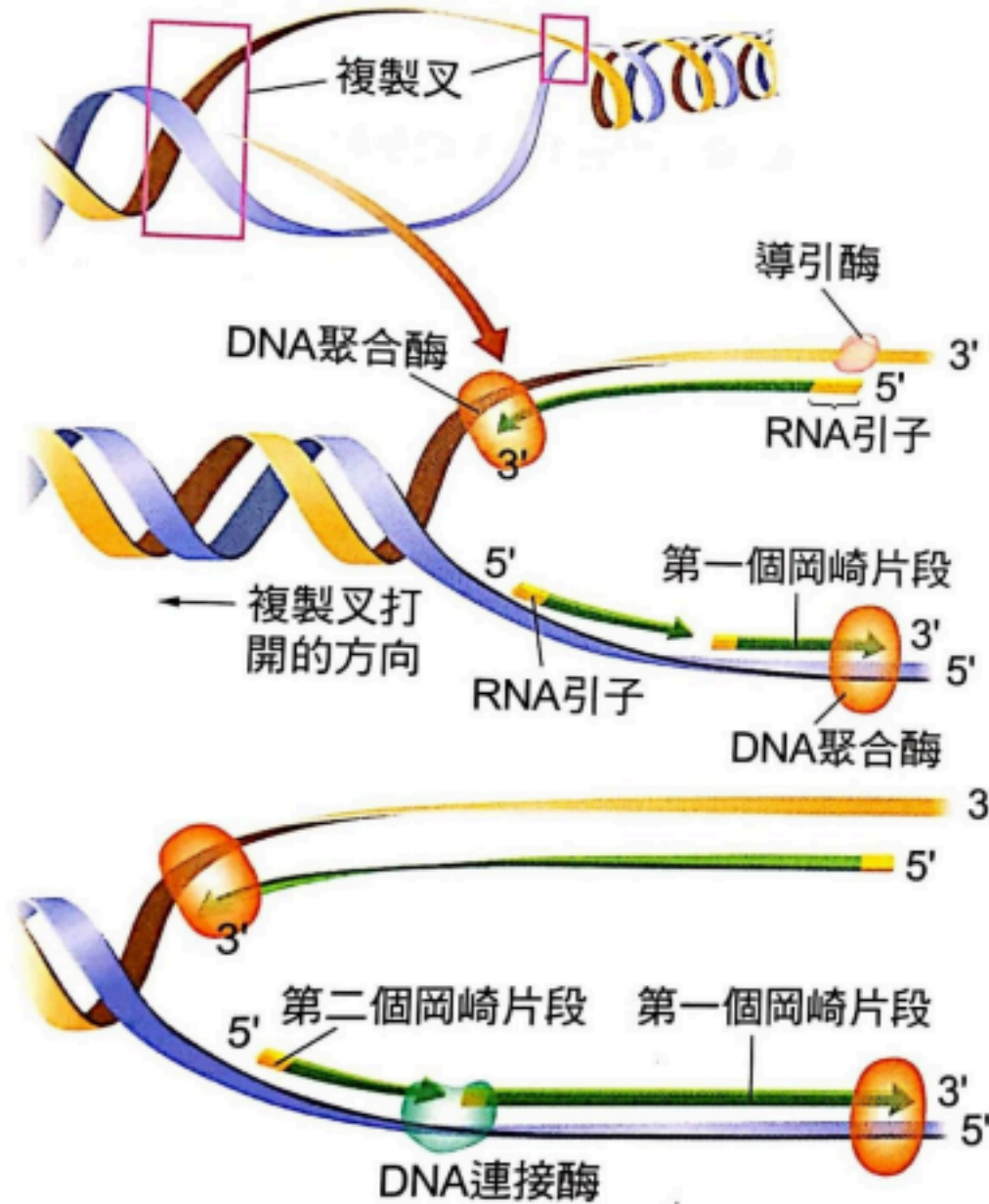
DNA分子的复制
(Replication of DNA)
~半保留式复制



(semiconservative replication)

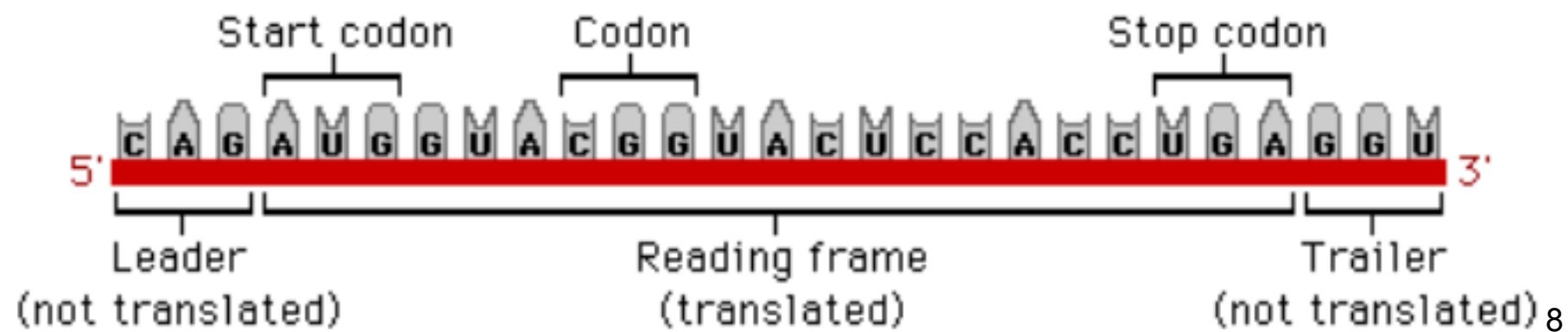


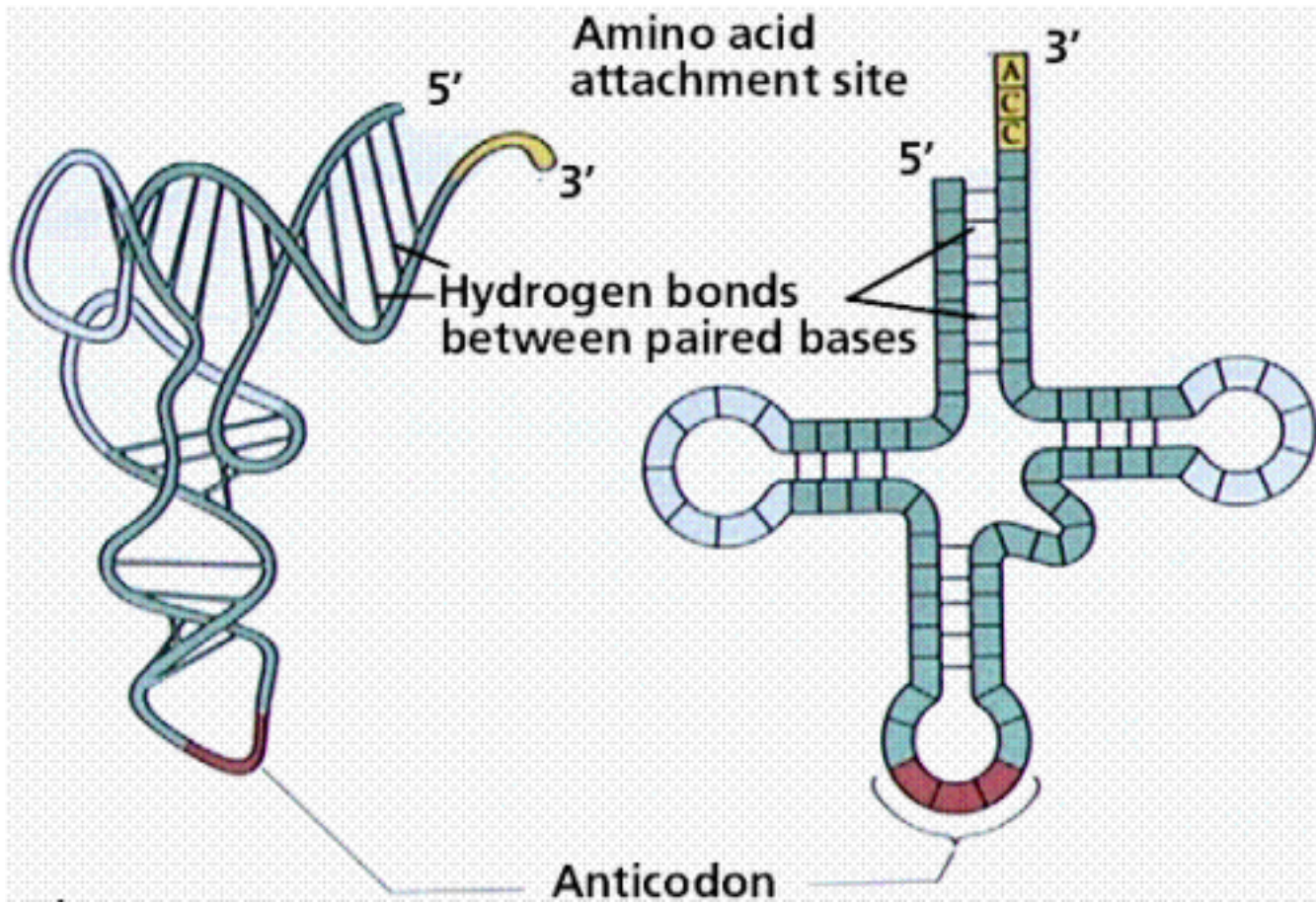
半保留式复制有什么意义? 甲 乙 丙 丁₅



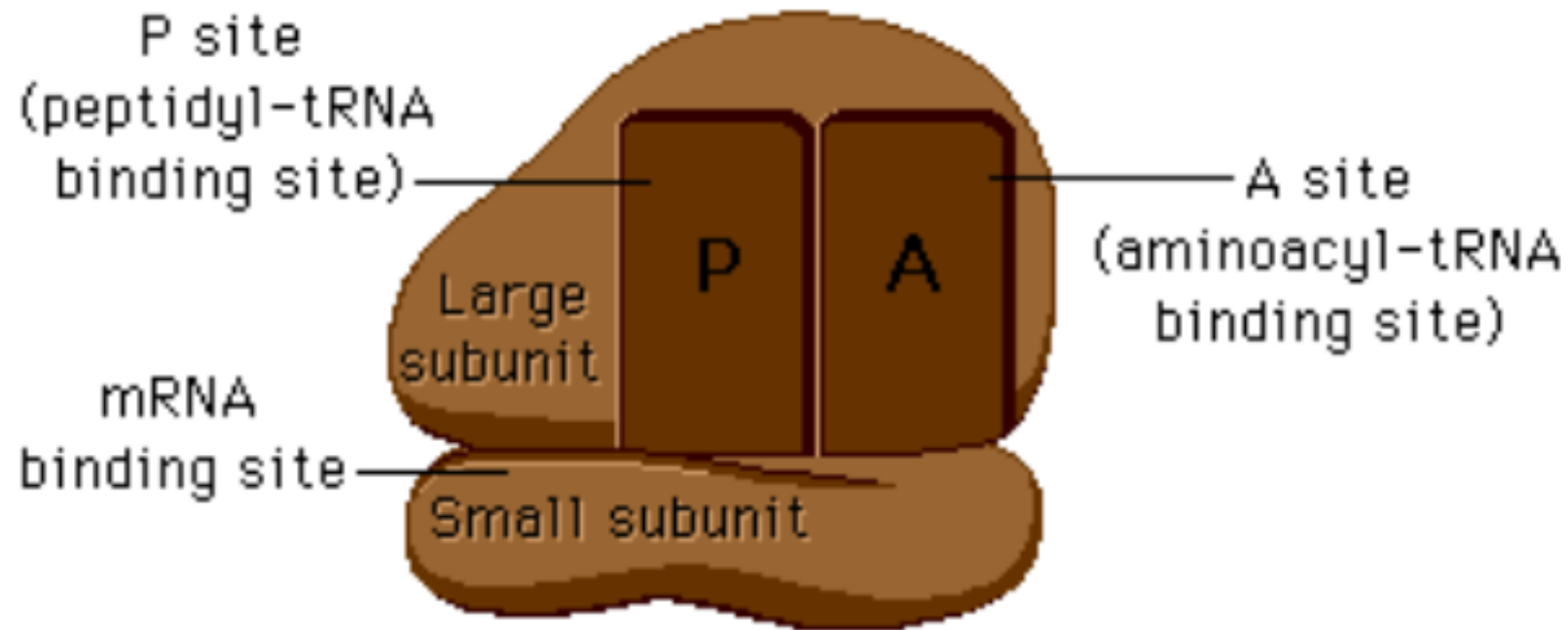
◀ DNA 聚合酶需要引子的存在才能沿 $5' \rightarrow 3'$ 的方向延伸新股。留意圖上方新股（綠色）之合成是連續的，下方新股（綠色）之合成先是不連續的，最後才由 DNA 連接酶把各個岡崎片段串連起來。

3种类、形态、功能



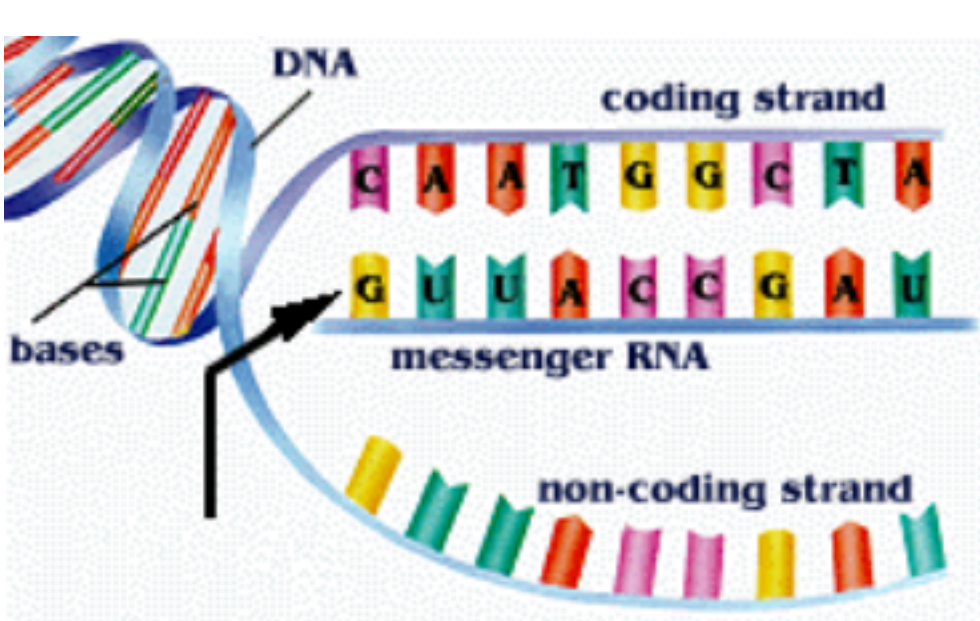


反密码子



10

3来源 ~轉录(transcription)



起始(initiation)

延伸

(chain elongation)

终止(termination)

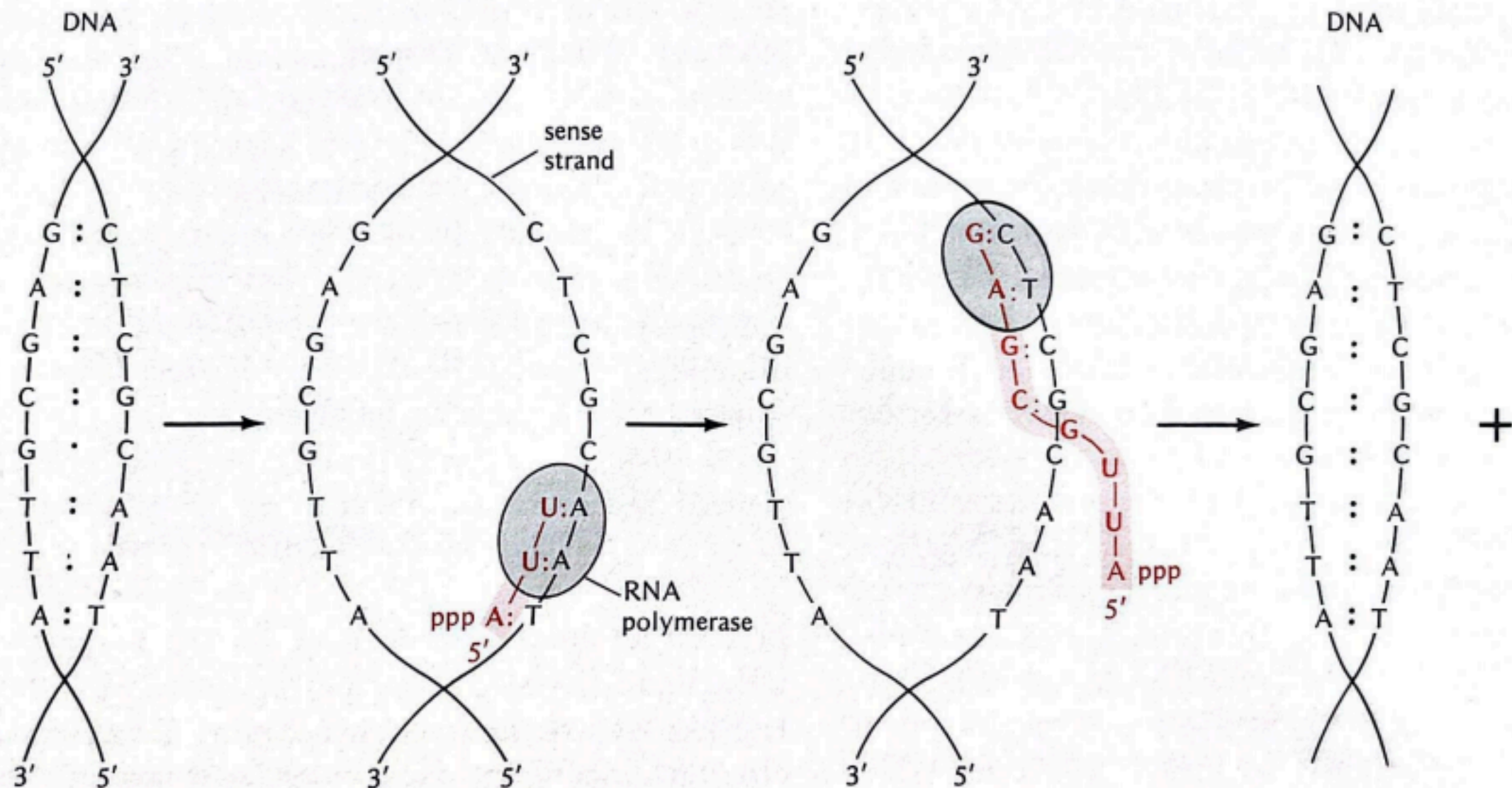


Figure 5-31

Model for the transcription of the nucleotide sequence from one strand of a DNA double helix (sense strand) into an RNA strand by DNA-directed RNA polymerase. In prokaryotes the RNA nucleotide at the 5' end is always a purine (A or G).

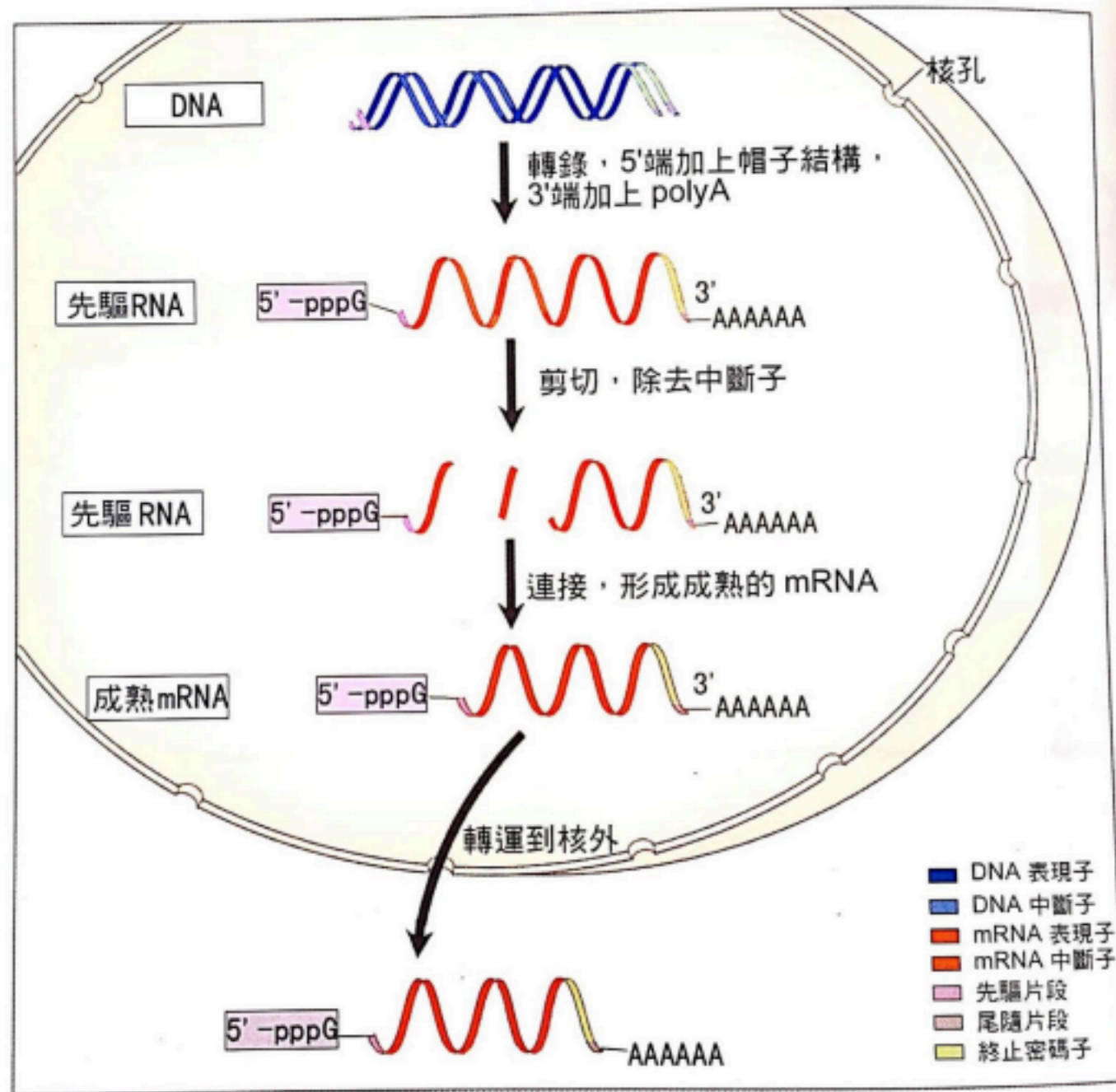
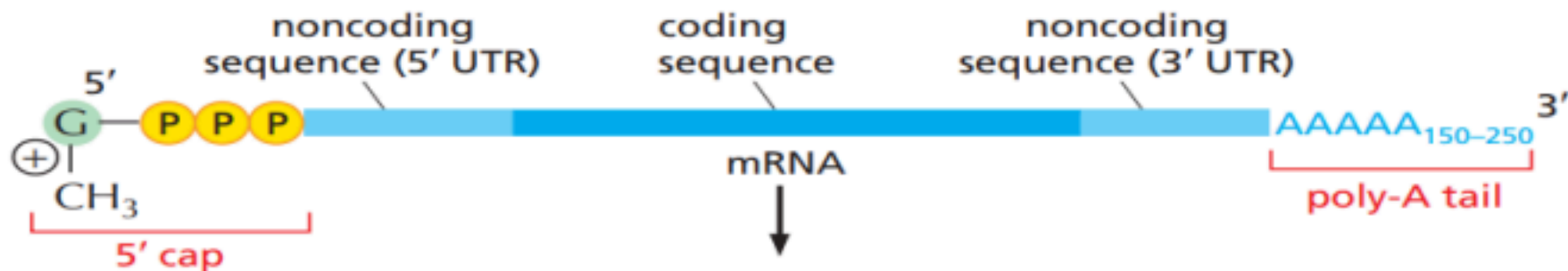


Figure 7–17 Eukaryotic mRNA molecules are modified by capping and polyadenylation. (A) A eukaryotic mRNA has a cap at the 5' end and a poly-A tail at the 3' end. In addition to the nucleotide sequences that code for protein, most mRNAs also contain extra, noncoding sequences, as shown. The noncoding portion at the 5' end is called the 5' untranslated region, or 5' UTR, and that at the 3' end is called the 3' UTR. (B) The structure of the 5' cap. Many eukaryotic mRNA caps carry an additional modification: the 2'-hydroxyl group on the second ribose sugar in the mRNA is methylated (not shown).

RNA capping and polyadenylation



翻译(translation)

Polysome/polyribosome

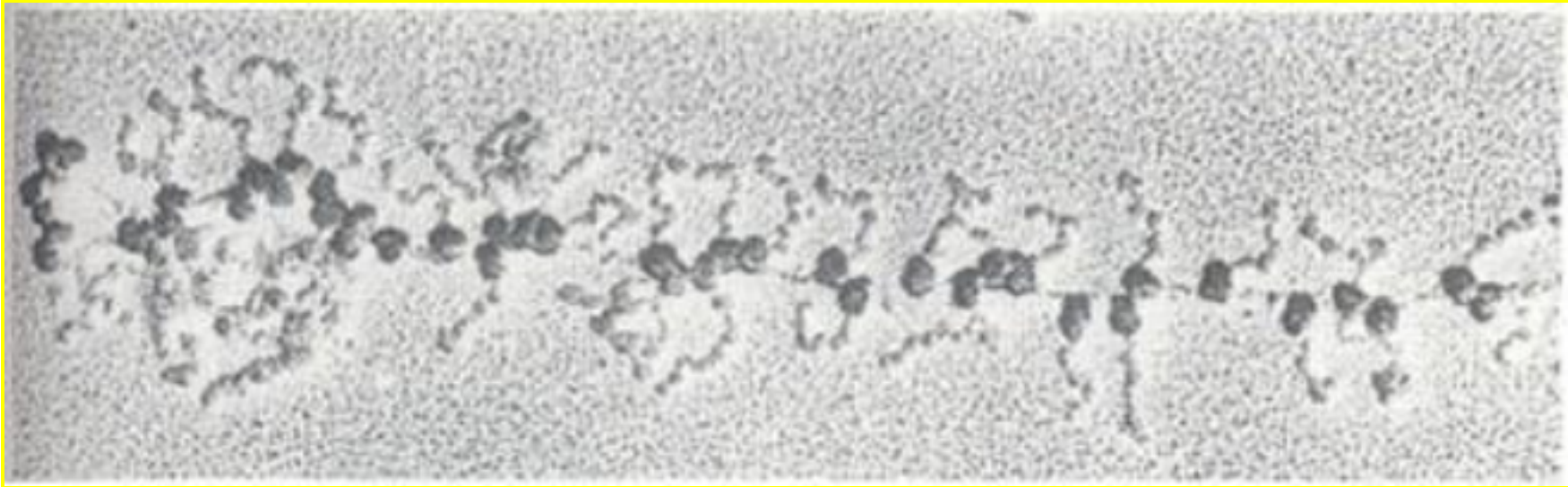
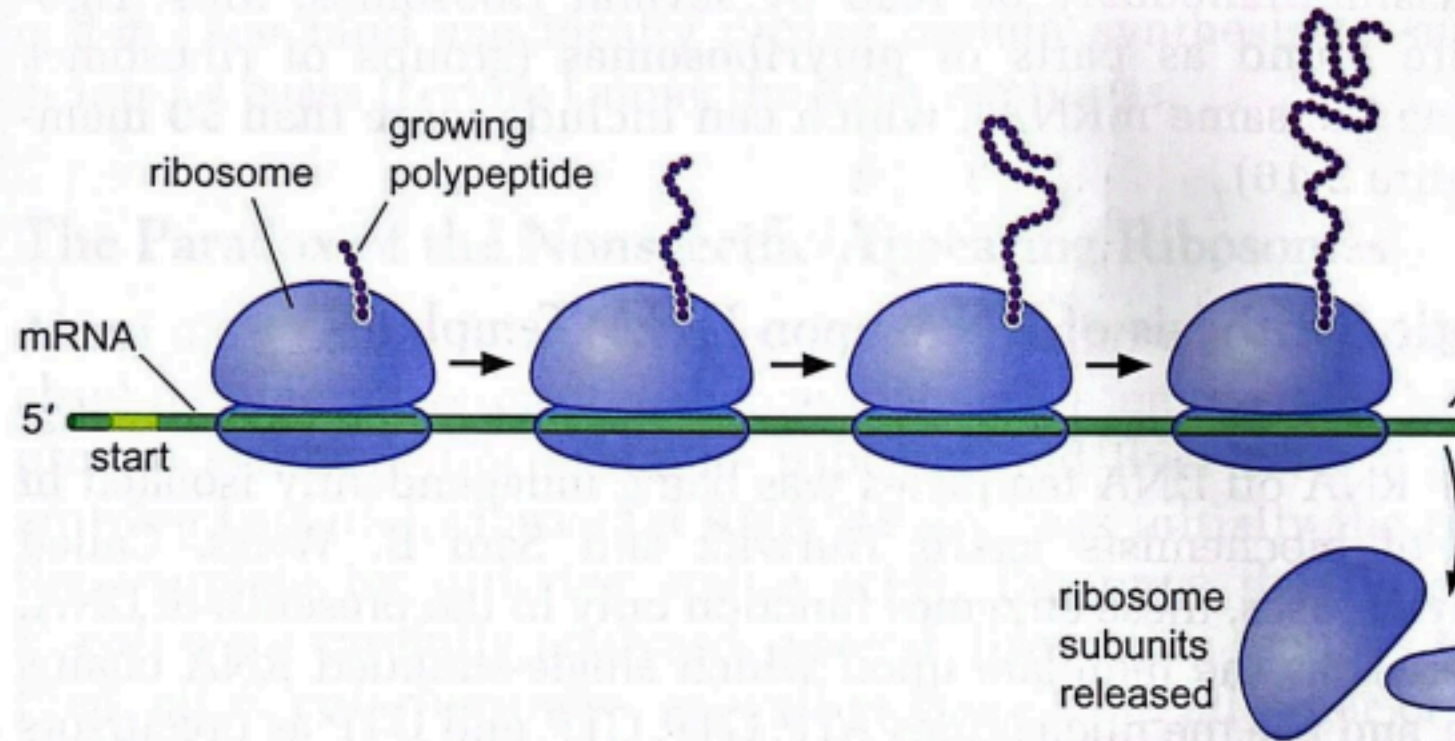


FIGURE 2-16 Diagram of a polyribosome. Each ribosome attaches at a start signal at the 5' end of an mRNA chain and synthesizes a polypeptide as it proceeds along the molecule. Several ribosomes may be attached to one mRNA molecule at one time; the entire assembly is called a polyribosome.



DNA与RNA的比较

比较项目 DNA RNA

全名
分布
基本组成单
位 硷基种类
五碳糖
结构
分类

--

DNA分子的复制、轉录、轉译的比较 复

制 轉录 轉译

发生时间

发生场所

模板

产物

硷基配对

遗传信息传递